Tennessee Eastman: benchmark signal reference

Paper section 3.13: *Tennessee Eastman process control problem*

12 manipulated variables: XMV

Valve and agitator commands are expressed as percentages.

1	D feed, stream 2
2	E feed, stream 3
3	A feed, stream 1
4	A/C mixed feed, stream 4
5	Compressor recycle bypass
6	Purge, stream 9
7	Separator liquid, stream 10
8	Stripper product, stream 11
9	Stripper steam
10	Reactor cooling-water outlet
11	Condenser cooling-water outlet
12	Reactor agitator command [%]

Source terminology sometimes calls the condenser coolant the separator coolant; it is the same thermal path. Commands and realized valve positions are distinct model quantities.

Representative disturbance paths

IDV(1): A/C feed ratio, constant B.
IDV(4): reactor coolant inlet temperature.
IDV(6): loss of A feed.
IDV(13): kinetics drift.
IDV(14–15): coolant valve sticking.

The repository's archived native runs cover nominal operation, IDV(1) and IDV(4); this diagram adds no new execution evidence.

Continuous measurements: XMEAS(1:22)

1	A feed	kscmh
2	D feed	kg/h
3	E feed	kg/h
4	A/C mixed feed	kscmh
5	Recycle flow	kscmh
6	Reactor feed	kscmh
7	Reactor pressure	kPa gauge
8	Reactor level	%
9	Reactor temperature	°C
10	Purge flow	kscmh
11	Separator temperature	°C
12	Separator level	%
13	Separator pressure	kPa gauge
14	Separator underflow	m^3/h
15	Stripper level	%
16	Stripper pressure	kPa gauge
17	Stripper underflow	m^3/h
18	Stripper temperature	°C
19	Stripper steam	kg/h
20	Compressor power	kW
21	Reactor coolant outlet	°C
22	Condenser coolant outlet	°C

kscmh: thousands of standard cubic meters per hour.

From the flowsheet to a learning or diagnostic experiment

Record the simulator revision, operating mode, solver, time base, random seed, initial state, command limits and disturbance schedule. Select measured channels before fitting or tuning. Respect the chronological split and analyzer timing.

Markov/HMM sensor modes in the course are explicitly synthetic overlays. They must not be identified with official process disturbances or with experimentally calibrated equipment-failure probabilities.

Composition analyzers: XMEAS(23:41)

All concentrations are molar percentages.

Reactor feed, stream 6
$23{:}28 \longleftrightarrow A, B, C, D, E, F$

Purge, stream 9
$29{:}36 \longleftrightarrow A, B, C, D, E, F, G, H$

Product, stream 11
$37{:}41 \longleftrightarrow D, E, F, G, H$

Analyzer sample/hold, noise and delay are part of the simulator. A composition signal is not an instantaneous complete state observation.

Verify the implemented state count

The survey describes 50 original states. The archived native C revision also uses `NX=50`, with 50 derivatives. Its introductory comment says 52, but that comment conflicts with the executable declarations. See the state-count audit; extra diagnostic outputs are not additional continuous states.

Signal labels and units checked against the Ricker/Bathelt archive's documented C interface. State count checked against its executable declarations. The gallery links the archive, source hash and state-count audit.

Original explanatory drawing. Source: Kroll & Schulte (2014), DOI: 10.1016/j.asoc.2014.08.034. Model variants are identified in the accompanying case sheet.